

COS80023 - Big Data

Learning Summary Report

Student Name: Arun Ragavendhar Arunachalam Palaniyappan

Student ID: 104837257

Tutor: Dr. Afzaal Hassan

Class: Thursday 14:30-16:30

Self-Assessment Details

The following checklists provide an overview of my self-assessment for this unit.

	Pass (P)	Credit (C)	Distinction (D)	High Distinction (HD)
Self-Assessment (please tick)				✓

Self-assessment Statement

	Included (please tick)
Learning Summary Report	✓
Pass tasks are marked as correct on Canvas	✓
Credit tasks are marked as correct on Canvas (optional)	✓
Distinction/High Distinction plan has been approved on Canvas (optional)	✓
Distinction/High Distinction report has been submitted on Canvas (optional)	✓
Distinction/High Distinction interview has been booked (optional)	✓

Submission Checklist

Declaration

I declare that this portfolio is my individual work. I have not copied from any other student's work or from any other source except where due acknowledgment is made explicitly in the text, nor has any part of this submission been written for me by another person.

Signature: Arun Ragavendhar

Introduction

I am writing this report to summarise what I learned in COS80023 Big Data. It includes my self-assessment against the unit learning outcomes, a short note on why I chose each piece of evidence, a clear account of how my work meets each outcome, and a reflection on my learning. My evidence spans all lectures, ten weekly labs (Weeks 1–10), and a Distinction/High-Distinction exploratory data analysis project in R on Victorian crash data. Together, these show end-to-end skills: setting questions, preparing and joining multi-table data, querying different storage models, building and checking visuals and statistics, and turning findings into practical actions.

Coverage of the Unit Learning Outcomes

This section explains how the included work shows my depth of understanding for each intended learning outcome of the unit.

ULO 1: Challenges and Opportunities of Big Data

Explain Big Data's hurdles and advantages, covering computing needs, privacy, and security, and relate these to what organisations can achieve with it.

What I learned: I can explain both sides. Opportunities: better decisions, faster services, new value from combining sources. Challenges: storage and extract, multi-format integration, data quality, privacy, etc. I paired each challenge with a practical response (cloud/distributed storage, simple ETL, validation rules, clear retention, measured cost controls).

My evidence:

1. Lab 1: Pass Task 1, Q1 (Opportunities).

COS80023 Big Data – Lab 1

Lab 1: Pass Task 1 – Opportunities and Challenges

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4. Explain how social media contributes to the opportunities of Big Data

Data types

Social media produces large amounts of data from posts, comments, likes, and shares. This helps organisations learn about people's interests, habits, and networks.

Targeted Marketing

When combined with other data sources, it can be used for targeted marketing. For example, knowing someone's birthday is coming and that their family member is expecting a baby allows ads for gifts and baby products to be shown.

Public Opinion, Trend Tracking and Brand Monitoring

Social media data can also track public opinion, trends, and customer feedback. It is used for brand monitoring and market research.

It also helps organisations test how audiences react to products or campaigns by analysing engagement and feedback.

Real Time Insights

Since social media is constantly updated, it is a continuous source for Big Data analysis. It also provides real-time insight into emerging topics, allowing organisations to respond quickly to public conversations.

2. Lab 1: Pass Task 1, Q2 (Challenges + how I would handle them).

2. Finding useful information in big data has its challenges. Name 5 challenges and explain how organisations can address them.

One challenge is storing large amounts of data while keeping it secure, fast, and able to grow. Using cloud systems and distributed storage can solve this.

Another challenge is combining data from different formats such as text, images, and numbers. ETL tools and data integration processes can align these formats.

A third challenge is interpreting data. Words may have similar meanings, different meanings, or spelling mistakes. Machine learning and natural language processing can help correct and interpret them.

A fourth challenge is deciding what to keep. Keeping too much takes up space, increases costs, and may create privacy risks. Clear rules for filtering and removing data are needed. The bulk of the data also makes it hard to process everything quickly.

A fifth challenge is protecting privacy and following laws like GDPR. This includes collecting minimal data, using it only for the original purpose, de-identifying personal information, keeping data in the same region, giving people access to their data, and reporting breaches quickly.

Poor data quality is another common challenge, as incomplete or inaccurate data can lead to wrong conclusions. Organisations can address this by setting up data validation checks and regular quality audits.

3. Lecture 1: slide titled “Why are people interested in it?”

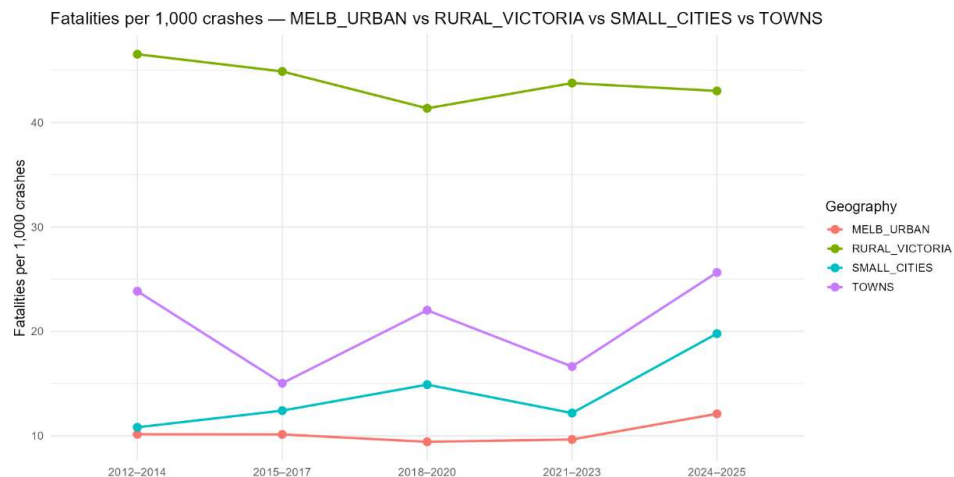
WHY ARE PEOPLE INTERESTED IN IT?

Example

- **Information**
 - We can collect information about our customers to better understand what they want.
 - Lucy has just bought a motorbike. Maybe she wants insurance, or boots, or a jacket.
- **“360° view”**
 - We can combine data about a customer to know better what they want.
 - According to her Facebook page, Lucy plans a road trip. She needs pannier bags.
- **Automation**
 - We can use programs to do useful things with this information.
 - An algorithm can work out what Lucy is up to and mail her ads about pannier bags.

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4. Final Project: Figure: “Exposure (share %) vs severity (per 1,000) by speed.”



How I applied it: In the D/HD project, I took nine tables which had road accidents related data, kept the 'crashes' as the unit of measurement, joined the tables on ACCIDENT_NO key, grouped by five different time blocks and four geographical regions, and reported fatalities per 1,000 crashes with exposure share. This produced an actionable finding: speed in rural roads with ≥ 90 km/h and mid-blocks and T-intersections carry the highest fatal risk.

Why this meets ULO 1: I show the organisational potential (turn raw data into decisions) and the practical hurdles (scale, heterogeneity, quality, privacy), with evidence from labs, lectures, and a full project where I applied those trade-offs end-to-end.

ULO 2: Storing and Managing Data

Suggest practical, efficient ways to store and manage very large datasets.

What I learned: I learned to choose the right store for the job (cloud SQL, HDFS/Blob), design stable keys and tables so that joins work, and manage the data on the HD clusters.

My evidence:

1. Lab 2: Azure SQL Query Editor: portal view with my database/tables and the join output listing SupplierName, Country, ProductName, Unit, Price.

The screenshot shows the Microsoft Azure portal interface for the SQL Query Editor. The browser address bar displays the URL: <https://portal.azure.com/#@swin.edu.au/resource/subscriptions/8bcd69b-9218-4a85-9916-dc4997da1db/resourceGroups/104837257...>. The page title is "104837257rdb (104837257server/104837257rdb) | Query editor (preview)".

On the left, the "Tables" list includes: dbo.Categories, dbo.Customers, dbo.Employees, dbo.OrderDetails, **dbo.Orders**, dbo.Products, dbo.Shippers, and dbo.Suppliers. The "dbo.Orders" table is selected.

The "Query 3" editor shows the following SQL query:

```
1 SELECT SupplierName, Country, ProductName, Unit, Price
2 FROM Suppliers s JOIN Products p
3 ON s.SupplierID = p.SupplierID
4 ORDER BY SupplierName ASC;
```

The "Results" tab displays the output of the query:

SupplierName	Country	ProductName	Unit	Price
Aux joyeux ecclésiastiques	France	Côte de Blaye	12 - 75 cl bottles	264
Aux joyeux ecclésiastiques	France	Chartreuse verte	750 cc per bottle	18
Bigfoot Breweries	USA	Sasquatch Ale	24 - 12 oz bottles	14
Bigfoot Breweries	USA	Steeleye Stout	24 - 12 oz bottles	18
Bigfoot Breweries	USA	Laughing Lumberjack Lager	24 - 12 oz bottles	14

A status bar at the bottom indicates "Query succeeded | 0s".

2. Lab 4: HDInsight + WASB/ADLS.

```
sshuser@hn0-s10483:~$ hdfs dfs -cat /output/part-r-00000
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/hdp/4.1.20.5/hadoop/lib/slf4j-reload4j-1.7.35.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/hdp/4.1.20.5/hadoop-hdfs/lib/slf4j-reload4j-1.7.35.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Reload4jLoggerFactory]
|=
#1 1
#2 1
#3 1
#4 1
#5 1
#include 1
&& 2
** 1
(python) 1
(data 3
(denom 2
(expected 1
(float 1
(for 1
(idx 2
(p 4
(staticFive 2
(summary) 5
(uint64_t) 1
(unsafe). 1
(username,)) 1
("u 1
* 2
*/ 5
*p 1
*p); 2
```

How I applied it: For relational data, I used Azure SQL to create tables and verify joins in the portal. For distributed files, I showed how HDInsight reads/writes via WASB/ADLS, mapping input and output paths clearly.

Why this meets ULO 2: I proposed and used practical storage options (cloud SQL, Blob/HDFS) and demonstrated data storage and management patterns with verifiable screenshots that show set-up, data stored, and results.

ULO 3: Information Retrieval

Use the right tools to query different kinds of data stores, including NoSQL systems.

What I learned: I learned to pull data from different stores and formats, streams (Kafka), and multi-table files (CSV in a project folder) in the final project work on the accidents data and use the right tools to turn them into usable results (CLI consumers for streams; R/dplyr for linked tables).

My evidence:

1. Lab 6: Kafka/Zookeepers: terminal screenshots showing producer terminal (typed messages), and consumer terminal (received messages).

```
PS C:\Users\arunr\Pictures\kafka_2.12-2.3.0> bin\windows\kafka-console-p
roducer.bat --broker-list 127.0.0.1:9092 --topic streams-plaintext-input

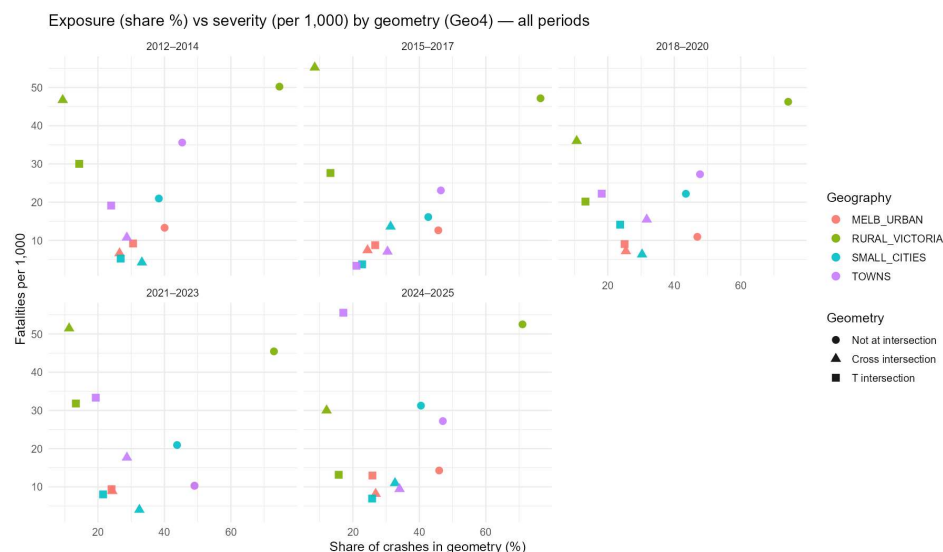
>Hi Kafka this
>is a test writing
>to check if the
>program works
>running successfully
>finished fully finished
>hello world hello
>
```

```

PS C:\Users\arunr\Pictures\kafka_2.12-2.3.0> bin\windows\kafka-console-c
onsumer.bat --bootstrap-server 127.0.0.1:9092 --topic streams-wordcount-
output --from-beginning --property print.key=true --property value.deser
ializer=org.apache.kafka.common.serialization.LongDeserializer
hi 1
kafka 1
this 1
is 1
a 1
test 1
writing 1
to 1
check 1
if 1
the 1
program 1
works 1
running 1
successfully 1
finished 1
fully 1
finished 2
hello 1
world 1
hello 2

```

- Exposure (share %) vs severity (per 1000) by road geometry, part of the works done regarding information retrieval, filtering, aggregation and analysis in the final project.



How I applied it: From the stream side, I retrieved live data by creating Kafka topics, sending messages with the producer, and reading them with the consumer CLI to confirm end-to-end flow. In the exploratory data analysis for the final project, I retrieved and joined six CSV tables on **ACCIDENT_NO** in R, filtered to five different time blocks and four regions, and queried the combined data with dplyr to produce simple, comparable tables (counts, fatalities, fatalities per 1,000, exposure share) that feed the figures.

Why this meets ULO 3: I used efficient tools for each store: Kafka CLI to retrieve records from a stream, and R/dplyr to query linked file tables at scale. The project shows I can turn raw, multi-source input into precise answers (e.g., where and when risk is highest), and Lab 6 shows I can retrieve information from a non-relational streaming system.

ULO 4: ETL

Choose and apply suitable technologies and methods to extract data and combine it from sources with different levels of structure.

What I learned: I learned how to take data from different places (CSV files and a small stream) and turn it into clean tables I can reuse. I kept the raw data, fixed types and missing values, removed duplicates, and saved tidy outputs.

My evidence:

1. Lab 5 (Pig, CSV to tidy output): I loaded a CSV, filtered out headers/blank rows, picked the columns I needed, grouped by a key (like country or payment type), added simple totals, sorted the results, and saved them to an output folder.

```
2025-09-04 05:45:56,564 [main] INFO  org.apache.hadoop.mapreduce.lib.in
put.FileInputFormat - Total input files to process : 1
2025-09-04 05:45:56,564 [main] INFO  org.apache.pig.backend.hadoop.exec
utionengine.util.MapRedUtil - Total input paths to process : 1
(Visa,836350)
(Mastercard,458450)
(Amex,188900)
(Diners,133800)
grunt>
```

2. Final project (crash data, end-to-end): I brought in the nine CSV tables, joined them by ACCIDENT_NO, fixed labels (lighting/surface/weather), removed duplicate records by keeping the newest, and saved a final table by year and region so charts load fast.

How I applied it: I first saved the raw files as it is, then made a cleaned copy with proper types and labels and finally wrote an analysis-ready table. I checked counts after each step and made sure joins on ACCIDENT_NO didn't lose rows.

Why this meets ULO 4: I compared simple tools (Pig vs my project steps) and applied them to extract and combine data with different shapes into one clean, reliable table that supports the analysis.

ULO 5: Machine Learning

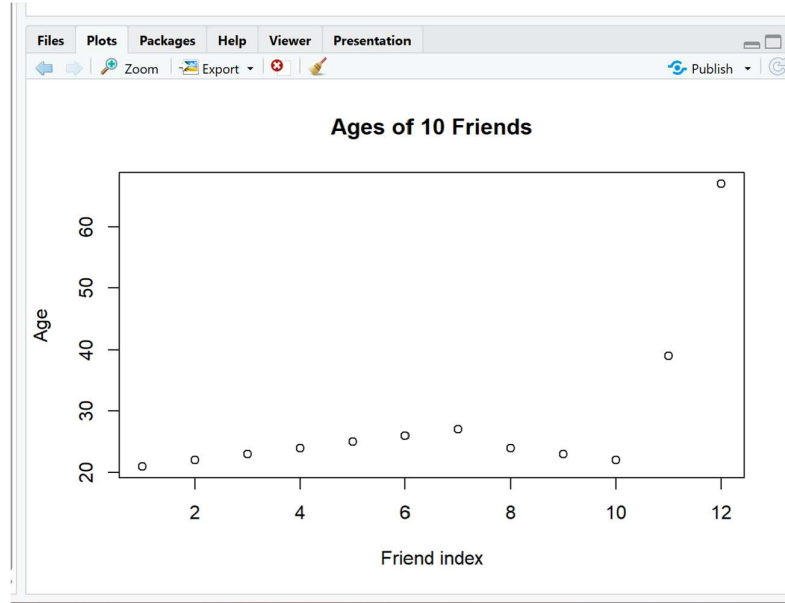
Judge whether the extracted information is reliable and useful by applying appropriate statistical checks.

What I learned: I learned to judge if results are trustworthy and useful by using simple checks: split the data into train/test, look at a confusion matrix, and read accuracy, sensitivity, and specificity together (not just one number). I also learned to watch for overfitting and use cross-validation to confirm the model holds up on new data.

My evidence:

1. Lab 7 (basics of ML): I practised simple stats first, mean, spread, plots, and then the key ML ideas: the difference between supervised and unsupervised work, how overfitting shows up

(good on train, worse on test), and how cross-validation gives a fairer score than a single split.



- Lab 8 (classifiers): I trained two small models (SVM and Random Forest) on iris and compared them using the same train/test split. I checked the confusion matrix per class and picked the better model based on test results, not just train results.

```
> confusionMatrix(table(svmresult, test$species))
Confusion Matrix and Statistics
```

svmresult \ test\$species	setosa	versicolor	virginica
setosa	15	0	0
versicolor	0	13	1
virginica	0	2	14

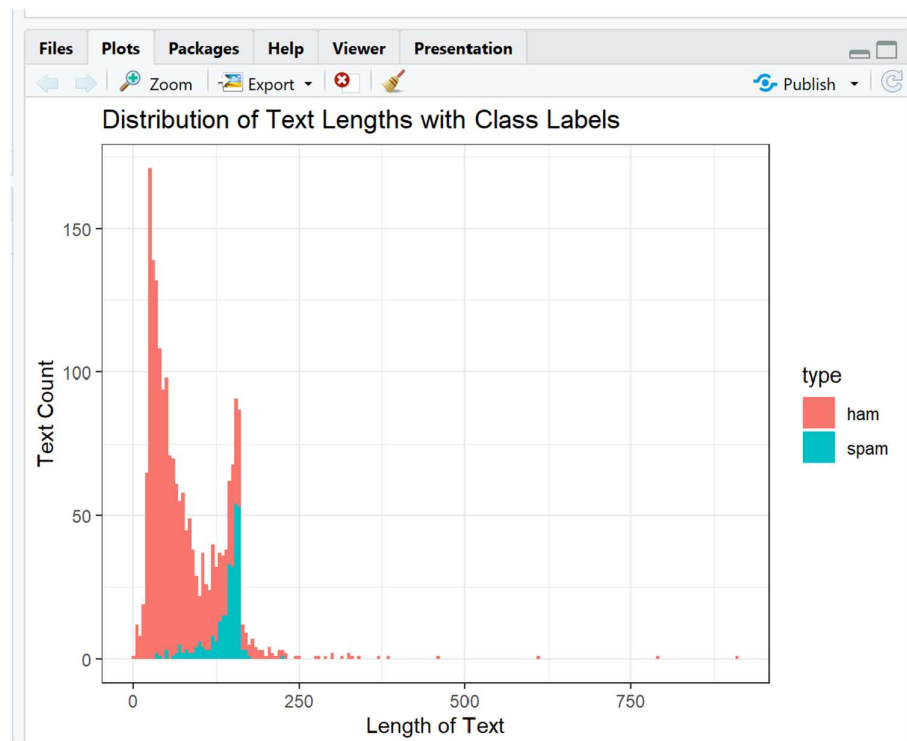
```
Overall Statistics

          Accuracy : 0.9333
          95% CI   : (0.8173, 0.986)
No Information Rate : 0.3333
P-Value [Acc > NIR] : < 2.2e-16

          Kappa    : 0.9

McNemar's Test P-Value : NA
```

- Lab 9 (NLP spam filter): I cleaned text, built simple word features, and trained a model to tell ham from spam. Because the classes were uneven, I checked sensitivity and specificity, not just accuracy, and noted where the model missed spam so I could improve it (features or threshold).



```
> confusionMatrix(table(trainresult, train.tokens.df$type))
Confusion Matrix and Statistics
```

```
trainresult  ham spam
ham      1149   49
spam      55   147
```

```
Accuracy : 0.9257
95% CI : (0.9107, 0.9389)
No Information Rate : 0.86
P-Value [Acc > NIR] : 1.198e-14
```

```
Kappa : 0.6954
```

```
Mcnemar's Test P-Value : 0.6239
```

```
Sensitivity : 0.9543
Specificity : 0.7500
Pos Pred Value : 0.9591
Neg Pred Value : 0.7277
Prevalence : 0.8600
Detection Rate : 0.8207
Detection Prevalence : 0.8557
Balanced Accuracy : 0.8522
```

```
'Positive' Class : ham
```

How I applied it: For each lab I kept the same routine: make a clean train/test split, train more than one model, and compare them with the same metrics; check class-by-class results; and confirm with cross-validation when possible. If train and test were close, I trusted the model; if not, I simplified or tried a different one.

Why this meets ULO 5: I used simple, clear statistics to judge the quality of the results and chose models that were fit for purpose, fair on new data, balanced across classes, and explained with numbers anyone can read.

Reflection

The most important things I learned: I learned to start with a clear question, keep the raw data safe, and build small steps I can trust. In the project, I joined the crash tables by ACCIDENT_NO, checked counts at each step, and only then made charts. This simple routine, of going from questioning to joining to checking to showing (the excel tables), kept me honest and helped me find where rural roads and higher speeds made crashes deadlier, in my final project.

What helped me most: Short, hands-on runs helped the most. In Lab 6 (Kafka) I sent a few lines like ACCIDENT_NO=V12345, speed_zone=100 and watched the consumer receive the same lines immediately, streams finally “clicked.” In Lab 5 (Pig) I loaded a messy CSV, dropped header/blank rows, grouped by country and payment type, and saved tidy totals; cleaning first made every later step easier. Training two small models’ side by side (SVM vs Random Forest) and reading the confusion matrix made model choice feel real, not theoretical.

What I found challenging: Messy labels and near-duplicates were the hardest. In the crash data in the project, the same event could appear more than once or with slight changes. I fixed this by keeping the newest record (based on update time) and using simple label maps for lighting, surface, and weather. In ML, it was tempting to trust a high accuracy, but class imbalance (like spam vs ham) forced me to check misses and weigh the cost of false alarms vs misses.

What I found most interesting: I liked putting share of crashes and deaths per 1,000 crashes side by side. It showed where volume is high, but risk is normal, and where volume is low, but risk is extreme. That contrast led to practical ideas: focus on rural mid-blocks and T-intersections at higher speeds, especially where surface quality is poor.

What I learned well: I built a repeatable way of working: save raw files, make a clean copy with fixed types and labels, join with one key, and check counts after each step. For ML, I now follow a small routine: split train/test, try more than one model, read the confusion matrix per class, and use cross-validation when possible.

What I still need to work on: I need to explain uncertainty better. Instead of “the model is good,” I should say “it got these classes right and struggled here,” with a short note on likely reasons (e.g., very short spam lines). I also want to automate basic checks (row counts, missing joins) so I can run them with one command.

My progress in this unit: I started by following lab steps and ended by setting my own steps for the project. Early on I copied examples; later I adapted them to fit the crash data and saved a final year×region table so charts load fast. The biggest jump was moving from “does this code run?” to “does this result make sense, and can I prove it?”

How this will help me in the future: At work I can pull data from different places, clean it, and show a clear, fair result that others can repeat. I can compare simple models on the same test set, explain trade-offs in plain English, and link findings to action (like where to focus road-safety work).

If I did this unit again: I would set up folders and checks on day one, write shorter notes after each lab, and lock a simple naming scheme for tables and charts. I would also decide earlier which two models to compare and stick with them to save time.

Other reflections: Small, clear wins build trust. A working Kafka send/receive, a Pig group-by that matches hand totals, a confusion matrix anyone can read, and a final table that reproduces the chart, these steps turned a big topic into work I can deliver and explain.

Supporting evidence and screenshots for completing all task in this unit (both mandatory and optional):

Pass tasks are marked as correct on Canvas and Credit tasks are marked as correct on Canvas.

Grades for Arun Ragavendhar Arunachalam Palaniyappan

 Print
Grades

Course

Arrange by

2025-HS2-COS80023-Big ▾

Due date ▾

Apply

Name	Due	Submitted	Status	Score	
Pass Task 1 Pass Tasks	14 Aug by 23:59	14 Aug at 6:04		1 / 1	
Pass Task 2 Pass Tasks	21 Aug by 23:59	14 Aug at 15:47		1 / 1	
Pass Task 3 Pass Tasks	28 Aug by 23:59	24 Aug at 17:01		1 / 1	
Pass Task 4 Pass Tasks	4 Sep by 23:59	29 Aug at 8:37		1 / 1	
Pass Task 5 Pass Tasks	11 Sep by 23:59	7 Sep at 15:44		1 / 1	
Credit Task 6 (optional) Credit Tasks	25 Sep by 23:59	25 Sep at 15:04		1 / 1	
Pass Task 7 Pass Tasks	2 Oct by 23:59	2 Oct at 12:15		1 / 1	
Pass Task 8 Pass Tasks	9 Oct by 23:59	9 Oct at 11:25		1 / 1	
Pass Task 9 Pass Tasks	16 Oct by 23:59	16 Oct at 16:56		1 / 1	  2

Credit Task 10 (optional) Credit Tasks	23 Oct by 23:59	23 Oct at 14:09	1 / 1	!
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Distinction/High Distinction plan has been approved on Canvas

Distinction/High Distinction Project Plan D/HD Project	17 Oct by 23:59	16 Oct at 22:03	7 / 7
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Distinction/High Distinction report has been submitted on Canvas and interview has been booked (optional)

Appointment
D/HD Interview

11 Nov, 12:30 - 12:45

Source Calendar2025-HS2-COS80023-Big Data

LocationCollaborate Ultra

DetailsOnly one member to sign up for a time. All other members to attend that same time (check first if there is a slot booked by a team member).

Attendees

- Arun Ragavendhar Arunachalam Palaniyappan

Comments
Team: Arun Ragavendhar Arunachalam Palaniyappan - 104837257 Vikas Kadruvan - 104981442 Nikolaos Anestakis - 104014513

Unreserve